

Bernoulli Differential Equation

First Order Linear Differential Equations

If P and Q are continuous functions of x , then

$$\frac{dy}{dx} + P(x)y = Q(x)$$

is a first order linear differential equation.

Example

$$\frac{dy}{dx} + \underline{2x} y = \underline{2e^{-x^2}}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

First order linear differential equation

$$\frac{d y}{d x} + P(x) y = Q(x)$$

$$M dx + N dy = 0$$

$$P(x) y dx - Q(x) dx + dy = 0$$

$$\underbrace{[P(x) y - Q(x)]}_M dx + \underbrace{1}_N dy = 0$$

$$\frac{\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right)}{N} = \frac{P(x)}{1} = P(x)$$

$$I.F. \mu(x) = e^{\int P(x) dx}$$

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The first order Bernoulli differential equation can be written in the following standard form

$$\frac{d y}{d x} + P(x) y = Q(x) y^n$$

real number
 $n \neq 0$ **and** $n \neq 1$

Example

$$\frac{d y}{d x} + y = x y^5$$

- Named after Jacob Bernoulli who discussed it in 1695.



real number

$n \neq 0$ **and** $n \neq 1$

$$\frac{d y}{d x} + P(x) y = Q(x) y^n$$

Bernoulli Differential Equation

$$\frac{d y}{d x} + P(x) y = Q(x)$$

First order linear differential equation

➤ Here $P(x)$ and $Q(x)$ are **continuous** functions of x .

➤ $n \neq 0$ **and** $n \neq 1$, thus the differential equation is **nonlinear**.

$$\frac{d y}{d x} + P(x) y = Q(x) y^{\boxed{n}} \quad n = 0$$

Standard form First order
linear (y and y')

$$\frac{d y}{d x} + P(x) y = Q(x)$$

first order linear differential
equation

$$\frac{d y}{d x} + P(x) y = Q(x) y^{\boxed{n}} \quad n = 1$$

$$\frac{d y}{d x} + \textcircled{P(x) y} = \textcircled{Q(x) y} \quad \text{Variable separable}$$

$$\frac{d y}{d x} = \underline{(Q(x) - P(x))} y$$

Strategy of solving Bernoulli equation

Putting the nonlinear equation into linear form

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \longrightarrow \textcircled{1}$$

Goal $\longrightarrow$ Linear

Change the dependent variable from y to z

$$z = y^{1-n}$$

$$\frac{dz}{dx} = (1-n)y^{(1-n)-1} \frac{dy}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

$$\frac{dz}{dx} = (1-n)y^{(1-n)-1} \frac{dy}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

Multiply the above standard form by $(1-n)y^{-n}$

$$(1-n)y^{-n} \frac{dy}{dx} + (1-n)P(x) \underbrace{y y^{-n}}^z = (1-n)Q(x) y^n y^{-n}$$

$$\frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x)$$

$$\frac{dz}{dx} + (1-n)P(x)z = (1-n)Q(x) \quad \text{linear in } z$$



first order linear differential equation

- Can therefore be solved analytically using **integrating factor method**.

Integrating factor

$$\mu(x) = e^{\int (1-n)P(x)dx}$$

To solve a Bernoulli equation

1. **Substitute**

$$z = \frac{1}{y^{n-1}} = y^{1-n}$$

2. **Solve the resulting first order differential equation.**

3. **After solving for z , determine the solution to the original differential equation.**

Example

$$\frac{d y}{d x} - \frac{1}{x} y = x y^2, \quad \text{where } x > 0$$

$$\frac{d y}{d x} + \underbrace{P(x)} y = \underbrace{Q(x)} y^n$$

$$n = 2$$

$$P(x) = \frac{-1}{x} \qquad Q(x) = x$$

$$z = y^{1-n} = y^{1-2} = y^{-1}$$

$$\frac{dz}{dx} = (1-n) y^{-n} \frac{dy}{dx} = -y^{-2} \frac{dy}{dx}$$



$$\frac{dz}{dx} + \frac{1}{x} z = -x$$

linear in z

Integrating factor

$$\frac{dz}{dx} + \frac{1}{x}z = -x$$

$$IF = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$x \frac{dz}{dx} + z = -x^2$$
$$\frac{d}{dx}(xz) = -x^2$$

$$xz = -\int x^2 dx$$

$$xz = \frac{-x^3}{3} + C$$

Use $z = \frac{1}{y},$ $\frac{x}{y} = \frac{-x^3}{3} + C$

$$\frac{1}{y} = -\frac{x^2}{3} + \frac{C}{x}$$

Application

Stability of the solution

➤ Special case of Bernoulli's equation

Logistic differential equation

$$\frac{dy}{dx} = y(1 - y)$$

$$\frac{dy}{dx} \ominus y = \ominus y^2$$

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

$$P(x) = -1$$

$$Q(x) = -1$$

$$n = 2$$

$$z = y^{1-n} = y^{1-2} = y^{-1}$$

$$\frac{dz}{dx} = (1-n)y^{-n} \frac{dy}{dx} = -y^{-2} \frac{dy}{dx}$$



$$\frac{dz}{dx} + z = 1$$

which is linear

Integrating factor

$$\frac{dz}{dx} + z = 1$$

$$IF = e^{\int dx} = e^x$$

$$e^x \frac{dz}{dx} + ze^x = e^x$$

$$\frac{d}{dx} (ze^x) = e^x$$

$$ze^x = e^x + c$$

Put back, use

$$z = \frac{1}{y}$$

$$\frac{1}{y} = 1 + ce^{-x}$$

$$y = \frac{1}{1 + ce^{-x}}$$

$$\frac{dy}{dx} = y(1-y)$$

$$\frac{dy}{dx} = 0$$

$$y(1-y) = 0$$

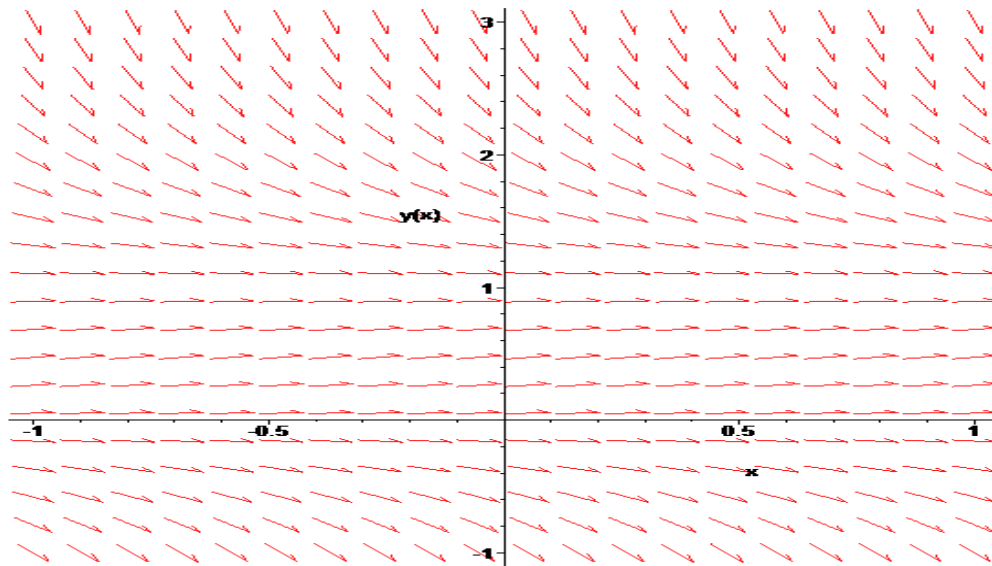
$$y = 0$$

$$(1-y) = 0$$

$$y < 0, \quad \frac{dy}{dx} \quad -$$

$$y > 1, \quad \frac{dy}{dx} \quad -$$

$$0 < y < 1, \quad \frac{dy}{dx} \quad +$$



$$y = 1$$

$$y = 0$$

Integrating factor

$$\frac{dz}{dx} + \frac{1}{x}z = -x$$

$$IF = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$x \frac{dz}{dx} + z = -x^2$$

$$\frac{d}{dx}(xz) = -x^2$$

$$xz = -\int x^2 dx$$

$$xz = \frac{-x^3}{3} + C$$

Use $z = \frac{1}{y}, \quad \frac{x}{y} = \frac{-x^3}{3} + C$

$$\frac{1}{y} = -\frac{x^2}{3} + \frac{C}{x}$$

$$\frac{dy}{dx} = y(1-y)$$

$$\frac{dy}{dx} = 0$$

$$y(1-y) = 0$$

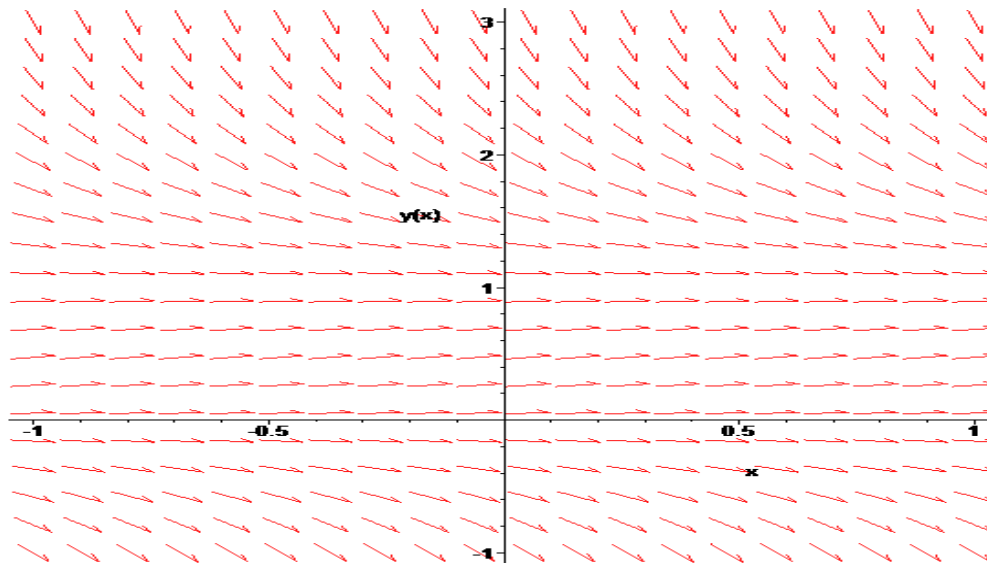
$$y = 0$$

$$(1-y) = 0$$

$$y < 0, \quad \frac{dy}{dx} \quad -$$

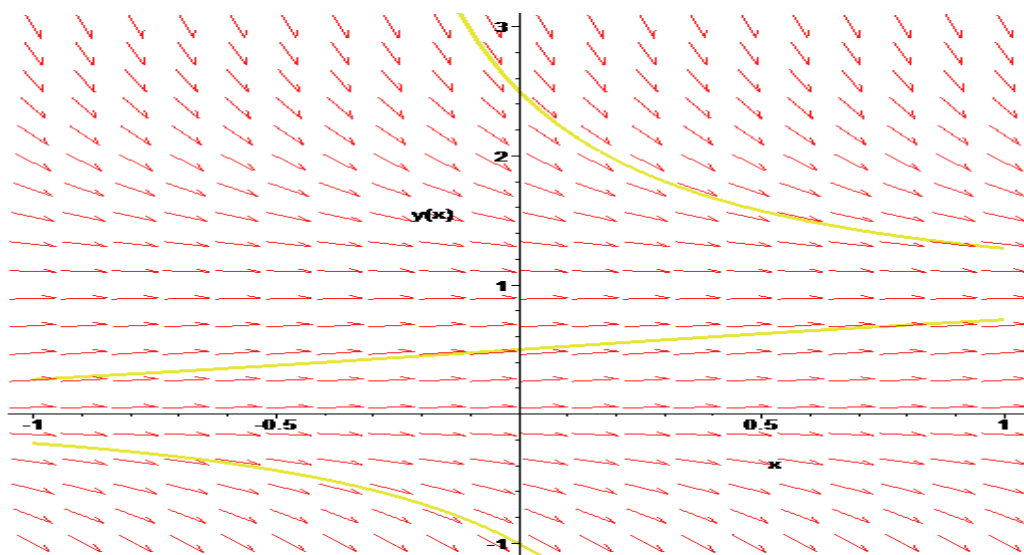
$$y > 1, \quad \frac{dy}{dx} \quad -$$

$$0 < y < 1, \quad \frac{dy}{dx} \quad +$$



$$y = 1$$

$$y = 0$$



$$y = 1$$

Stable

$$y = 0$$

Unstable